PANJAB UNIVERSITY CHANDIGARH

(Estted. under the Panjab University Act VII of 1947-enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

DISASTER MANAGEMENT
(ADD-ON COURSE)
CERTIFICATE AND DIPLOMA COURSE
EXAMINATION, 2013

PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and courses of reading for Certificate & Diploma Course in Disaster Management (Add on Course) for the examinations of 2013.

CERTIFICATE COURSE

Course Structure

Course Code	Title of the Paper	Max. Marks
CCDM I	Fundamentals of Disaster Management Theory : 75 Practical : 25	100
CCDM II	Disasters in India Theory : 75 Practical : 25	100

CCDM I: Fundamentals of Disaster Management

Maximum Marks : 100
Theory Paper : 75
Practical : 25
Time : 3 hrs.

Objectives: To introduce the students about basic concepts of various types of disasters and educate them on basic disaster management strategies.

Unit-I

- Definition and Concept of Hazard, Risk, Vulnerability and Disaster
- Types and classification of Disasters

Unit-II

- Nature Induced Disasters:

  Earthquakes, Floods and Cyclones.

  Avalanches, Forest Fire and Tsunami.

Unit-III

- Human Induced Disasters:
  Nuclear, Chemical and Industrial Disasters
  Global warming; Biological Disasters; Epidemics

Unit-IV

- Disaster Management: Meaning, Concepts, Approaches, Principles, Objectives and Scope
- Essentials of Disaster Management; Institutional and Individual's responsibilities during risk reduction, preparedness, response and recovery phases

Pedagogy:
Focus should be on observation, assessment and careful analysis of hazard prone areas to comprehend their vulnerability to disasters.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt one question from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical:

1. A Visit to a school for vulnerability assessment.
2. A Visit to any Government Hospital to know their disaster management plan
3. A visit to one slum area in the City to assess and analyze risk.

Note:

1. Based on these visits, students should prepare a report containing the vulnerability assessment of a village, disaster management plan of a hospital and risks of disasters in a slum.
2. The report should be submitted two weeks before the exam. The report will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

References

1. B.K.Khanna (2005): *Disasters: All You Wanted to Know About*, New India Publishing Agency, New Delhi.

2. Bryant Edwards (2005): *Natural Hazards*, Cambridge University Press, U.K.

3. Chakraborty, S.C.(2007): **Natural** *Hazards and Disaster Management*, Pragatishil Prokashak, Kolkata.

4. Sahni, Pardeep et.al. (eds.) 2002, *Disaster Mitigation Experiences and Reflections*, Prentice Hall of India, New Delhi.

CCDM II: Disasters in India

Maximum Marks	: 100
Theory Paper	: 75
Practical	: 25
Time	: 3 hrs.

Objectives: To introduce the students about the various disasters that have occurred in the recent past and their management in India.

Unit-I

- Regional Profile of India based on Earthquakes, Droughts, Floods and Cyclones.

Unit-II

- Disaster Management Frame in India
- National Level, including National Disaster Management Authority.
- State Authorities, Local Groups and Committees

Unit-III

- Case Studies: Bhopal Gas Tragedy, 1984; Orissa Super Cyclone, 1999; Kashmir Earthquake, 2005, and Kosi River Flood, 2009

Unit-IV

- Role of NGOs, Charitable Trusts, Army and Police, and Educational Institutions in Disaster Management

Pedagogy:

Visuals and DVDs on various disasters in India shall be shown to students.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt one question from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical Exercises: A field visit of the students will be organized by the College to some nearby site of any recent disaster/ event or ecological destruction or reconstruction and each student would prepare a Field Report based on his/her experiences.

Note: The Field Work based report of 25-30 pages prepared by the students will be submitted two weeks before the theory examination. This will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

References:

1 Kapur, A.: *Disasters in India: Studies of Grim Reality*, Rawat Publications, Jaipur; 2005.

2 Paraswamam, S. and Umikrishnan, P.V., *India Disaster Report*, Oxford University Press, New Delhi, 2000.

3 Reddy, K.R and Others eds. (2009) : *Natural Hazards and Disasters*, Department of Geography, Sri Krishnadeveraya University, Anantapur, A.P., India.

4 Sharma, V.K. ed. (1994): *Disaster Management*, Indian Institute of Public Administration, New Delhi.

DIPLOMA COURSE

Course Structure:

Course Code	Title of the Paper	Max. Marks
DCDM I	Remote Sensing in Disaster Management Theory : 75 Practical : 25	100
DCDM II	GIS in Disaster Management Theory : 75 Practical : 25	100

DCDM-I: Remote Sensing in Disaster Management

Maximum Marks : 100
Theory Paper : 75
Practical : 25
Time : 3 hrs.

Objectives: To introduce the students about the basic concepts and application of remote sensing in Disaster Management

Unit – I

- Basics of Remote Sensing:
  - Introduction, Definition, Nature, Development, and Limitations.
  - Role of Remote Sensing in Disaster Management

Unit – II

- Basics of GPS:
  - Concept, Definition, Elements and Factors of GPS.
  - The Mechanism of GPS for Disaster Management.

Unit – III

- Remote Sensing Applications in Natural Disasters:
  - Earthquake
  - Land Slides
  - Floods

Unit – IV

- Remote Sensing Applications in Human Induced Disaster:
  - Fire
  - Nuclear
  - Chemical

Pedagogy:

Basic Fundamentals of Remote Sensing and GPS to be introduced by Arial photograph interpretation. Training in all aspects of GIS & Remote Sensing and its real world applications.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt FOUR questions selecting one from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical :

Field Trips (one week practical attachment programme visit to Emergency sites, man-made and natural hazard locations)
To visit the sites of previous or current disaster as well as as agencies dealing with disasters or disaster prone areas/districts in various ways.

Note:

a) Based on these visits, students should prepare a report.
b) The report should be submitted two weeks before the exam. The report will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

Essential Readings:-

1. Natural Hazards and Disaster Management: R.B. Singh, Rawat Pulications-Jaipur, New Delhi, Banglore, Mumbai and Hydrabad.-2009.

2. Remote Sensing Applications in Applied Geosciences: Saumitra Mukherjee, Manak Publishers Pvt. Lid.-1999.

3. Remote Sensing and Geographical Information System: A.M. Chandra and S.K. Ghosh, Narosa Pulishing House, New Delhi, Chennai, Mumbai and Calcutta.-2006.

4. Integration of GIS and Remote Sensing: Victor Mesev, John Wiley and Sons Ltd.-2007.

5. Remote Sensing and Image Interpretation: Thomas M. Lillesand, Ralph W. Kiefer, Jonathan W. Chipman, Wiley India Pvt. Ltd. New Delhi.-2003.

6. Remote Sensing-Models and Methods for image Processing: Robert A. Schowengerdt, Academic Press- California U.S.A.-2006.

7. Remote sensing and GIS technologies for monitoring and prediction of disasters: Shailesh Nayak, Siyka Zlatanova, Springer A1Books.co.in Rediff Books Flipkart Infibeam.-2008.

8. Role of Remote Sensing in Disaster Management: Nirupama, Slobodan P. Simonovic, University of Western Ontario. Dept. of Civil and Environmental Engineering.-2002.

9. The SAGE handbook of remote sensing edited by: Giles M. Foody, Timothy A. Warner, M. Duane Nellis, SAGE Publications Ltd. Copyright.-2009.

10. Geomatics Solutions for Disaster Management: By Jonathan Li, Sisi Zlatanova, Andrea G. Fabbri, Springer A1Books.co.in Rediff Books Flipkart Infibeam.-2007.

DCDM-II: GIS in Disaster Management

Maximum Marks : 100
Theory Paper : 75
Practical : 25
Time : 3 hrs.

Objectives: To introduce the students about the basic concepts and application of GIS in Disaster Management.

Unit – I

- Basics of GIS:
  - Introduction, Definition, Nature, Development.
  - Components of GIS
  - Application of GIS in Disaster Management

Unit – II

- Geo-informatics Perspectives in Disaster Management:
  - Structure for Storing Disaster Data.
  - Spatial Analysis of Disaster.

Unit – III

- Disaster Mitigation Mapping:
  - Natural Hazard Analysis and Mapping
  - Man Made Hazards and Mapping

Unit – IV

- Risk and Vulnerability Assessment:
  - Warning and Evacuation
  - Consolidated Preparation for next Disaster

Pedagogy:

Basic Fundamentals of GIS to introduce Disaster Management.

Note:

a) There shall be one compulsory question containing 15 short type questions covering the whole syllabus. The student shall attempt any ten parts in about 25-30 words each. Each part shall carry 1.5 marks (15 marks).

b) The whole syllabus shall be divided into 4 units. Eight questions will be set from the whole syllabus, two from each unit. The student will be required to attempt FOUR questions selecting one from each unit. Each question will carry 15 marks. This will be in addition to the compulsory question.

Practical

Practical Disaster Exercises and online activities.

Note:

a) Based on these visits, students should prepare a report.
b) The report should be submitted two weeks before the exam. The report will be evaluated by a Board of Examiners consisting of the teacher teaching the course and one faculty member from the allied discipline.

Essential Readings:-

1. Introduction to Geographical Information System: Dr. M.A.Siddiqui, Sharda Pustak Bhawan Allahabad.-2005.

2. Integration of GIS and Remote Sensing: Victor Mesev, John Wiley and Sons Ltd.-2007.

3. GIS- Fundamentals, Applications and Implementations: Dr. K. Elangovan, New India Publishing Agency- Pitampura- New Delhi.-2006.

4. Introduction to Geographical Information System: Lan Heywood, Sarah Cornelius, Steve Carver, Srinivasa Raju, Dorling Kindersley India Pvt. Ltd.2006.

5. Introduction to Geographic Information Systems: Kang Tsung Chang, Tata Mc Graw-Hill, Publishing Company Ltd.- New Delhi.2006.

6. Greg Bankoff Frerks, Dorothea Hilhorst, Mapping Vulnerability: Disaster Development and People, Earthscan Publications Ltd.,2004.

7. Monmonier Mark: Cartographies of Danger: Mapping Hazards in America, Univ. of Chicago Press Publication.-1997.

8. Geo-information for disaster management: By Petrus Johannes Maria van Oosterom, Siyka Zlatanova, Elfriede M. Fendel, Springer A1Books.co.in Rediff Books Flipkart Infibeam-2005.

9. GIS for disaster management: proceedings: Hadeki Kaji, United nations centre for regional development, 1997.

10. Disaster damage estimation models: Data needs vs. ground reality: By Sudha Maheshwari, Rutgers The State University of New Jersey - New Brunswick-2007.

11. Frontiers of geographic information technology: Sanjay Rana, Jayant Sharma, Birkhäuser, 2006 - Science.

12. "Concepts and Techniques of Geographic Information Systems": C.P.Lo and Albert K.W.Yeung 2005 Prentice Hall of India, New Delhi.

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